

Case Study Scenario: Your Favourite Retail Chain Store Expansion

The Client: Your Favourite Retail Chain is a rapidly growing technology retail chain with a number of existing branches across the country and plans to open new outlets within 6 months. The company is facing a critical infrastructure bottleneck.

* You can give a unique name or pick any real-life example of this Client.

The Problem: Currently, the network engineering team manually configures every switch and router for new branch openings. This manual approach has led to inconsistent configurations, security vulnerabilities (such as open ports and misconfigured ACLs), and delayed store openings. The CTO has mandated a shift to Network Automation to standardize deployments and reduce human error.

Your Mission: As a team of Network Automation Engineers, you have been tasked with modernizing the infrastructure. You must demonstrate how software can be used to automate network and security provisioning, maximizing functionality and efficiency. Your goal is to move away from manual CLI interaction toward a programmable, Python-driven network environment.

This project involves a number of tasks:

1. Create a team of four members.
2. Python and Cisco network automation should be studied.
3. Present a brief proposal and demonstration regarding how Cisco network automation can be used to automate tasks for the Client:
 - (a) VLAN (multiple VLANs, access and trunk port configuration, inter-switch trunking)
 - (b) Multiaccess OSPFv2 Network
 - (c) ACLs,
 - (d) NAT,
 - (e) VPN
 - (f) IPSec
4. You are free to use any simulation tools such as Cisco Modeling Labs (CML), GNS3, EVE-NG, Docker, existing Virtual Machines, or others to show your demonstration. Minimum number of nodes for the simulation: **5 router/switch nodes** to simulate different sites of the Client branch.
5. Numerous YouTubers have posted content demonstrating how network automation can enhance computer network activities, such as [<https://www.youtube.com/watch?v=FX5m50xfFyM>]. Although you are permitted to

study all of them, your project cannot be identical to theirs. Even if the differences are slight, it should be distinct. The objective here is for you to be creative and produce original work, distinct from that of others.

6. Any creative solutions involving Network Automation is appreciated.
7. The items that must be submitted for project completion:
 - (a) A complete report of the proposal and demonstration (including ALL automation scripts used in the project).
 - (b) A presentation slide describing the project details.
8. The presentation is in Week 14, while submission deadline is due on Week 15, Thursday. The materials must be uploaded to Spectrum.

Total allocated marks for this Group Assignment are 15 marks of your overall grades.